

# The complete $\eta^{-2/3}$ strong-drive energy coefficient

Maciej Janowicz

## 1 Symbols and exact quadratic reduction

The symbolic calculation uses

$$n \in \mathbb{N}_0, \quad \delta \in \mathbb{R}, \quad \lambda = r^{-1} > 0, \quad \epsilon = \eta^{-1/3} > 0.$$

In particular, no sign assumption is imposed on the detuning  $\delta$ . After  $a = b - r$ ,  $r^3 + \delta r = \eta$ , direct normal ordering gives

$$\frac{H}{V} = -\frac{3}{2}r^4 - \delta r^2 + (2r^2 + \delta)b^\dagger b + \frac{r^2}{2}(b^{\dagger 2} + b^2) - r\mathcal{O}_3 + \mathcal{O}_4,$$

where

$$\mathcal{O}_3 = b^{\dagger 2}b + b^\dagger b^2, \quad \mathcal{O}_4 = \frac{1}{2}b^{\dagger 2}b^2.$$

For  $A = 2r^2 + \delta$  and  $\Omega = \sqrt{A^2 - r^4}$ , take  $b = C_\lambda c + S_\lambda c^\dagger$ ,

$$C_\lambda^2 = \frac{1}{2} \left( \frac{A}{\Omega} + 1 \right), \quad S_\lambda^2 = \frac{1}{2} \left( \frac{A}{\Omega} - 1 \right), \quad 2C_\lambda S_\lambda = -\frac{r^2}{\Omega}.$$

The chosen branch has  $C_\lambda > 0, S_\lambda < 0$ . The anomalous coefficient is

$$AC_\lambda S_\lambda + \frac{r^2}{2}(C_\lambda^2 + S_\lambda^2) = 0,$$

which directly audits the sign. If

$$C_0 = \sqrt{\frac{1}{2} + \frac{1}{\sqrt{3}}}, \quad S_0 = -\sqrt{-\frac{1}{2} + \frac{1}{\sqrt{3}}},$$

then

$$\begin{aligned} C_\lambda &= C_0 \left[ 1 + \delta \left( -\frac{1}{3} + \frac{\sqrt{3}}{6} \right) \lambda^2 + \delta^2 \left( \frac{17}{72} - \frac{\sqrt{3}}{9} \right) \lambda^4 + O(\lambda^6) \right], \\ S_\lambda &= S_0 \left[ 1 + \delta \left( -\frac{1}{3} - \frac{\sqrt{3}}{6} \right) \lambda^2 + \delta^2 \left( \frac{17}{72} + \frac{\sqrt{3}}{9} \right) \lambda^4 + O(\lambda^6) \right]. \end{aligned}$$

## 2 Recursive perturbation theory

Define

$$\begin{aligned} \mathcal{K}(\lambda) &= \lambda^2(H/V - E_{\text{vac}}^{(2)}) = \omega(\lambda)N - \lambda\mathcal{O}_3(\lambda) + \lambda^2\mathcal{O}_4(\lambda), \\ \omega(\lambda) &= \sqrt{3 + 4\delta\lambda^2 + \delta^2\lambda^4}. \end{aligned}$$

Expanding the complete operators gives  $\mathcal{K} = \sum_{j=0}^4 \lambda^j K_j + O(\lambda^5)$ . Thus  $K_3$  contains the  $\lambda^2$  correction to every factor in  $\mathcal{O}_3$ , and  $K_4$  contains the corresponding correction to every factor in  $\mathcal{O}_4$ , as well as  $-\delta^2 N/(6\sqrt{3})$ .

For intermediate normalization, put  $\psi^{(0)} = |n\rangle$ . Coefficient comparison in the eigenvalue equation gives, recursively,

$$\kappa_q = \left\langle n \left| \sum_{j=1}^q K_j \psi^{(q-j)} \right. \right\rangle,$$

$$\langle m | \psi^{(q)} \rangle = \frac{\left\langle m \left| \sum_{j=1}^q (K_j - \kappa_j) \psi^{(q-j)} \right. \right\rangle}{\sqrt{3}(n-m)}, \quad m \neq n.$$

Algebraic ladder action retains all reachable states and annihilates forbidden negative occupations. It gives

$$\kappa_1 = \kappa_3 = 0,$$

$$\kappa_2 = \frac{2\sqrt{3}}{3}\delta n - \frac{n^2 + n}{12} - \frac{1}{72},$$

and

$$\begin{aligned} \kappa_4 = & -\frac{\sqrt{3}}{18}\delta^2 n + \frac{\delta n^2}{18} - \frac{\sqrt{3}\delta n}{9} + \frac{\delta n}{18} - \frac{\sqrt{3}\delta}{18} + \frac{13\delta}{108} \\ & + \frac{5\sqrt{3}n^3}{1296} + \left(\frac{5\sqrt{3}}{864} - \frac{1}{18}\right)n^2 + \left(\frac{139\sqrt{3}}{2592} - \frac{1}{18}\right)n - \frac{5}{108} + \frac{67\sqrt{3}}{2592}. \end{aligned}$$

### 3 Independent finite-sum audit

The second calculation uses the generalized fourth-order finite formula directly. With  $D_k = \sqrt{3}(n-k)$  and  $(K_j)_{ab} = \langle a | K_j | b \rangle$ ,

$$\begin{aligned} \kappa_4 = & (K_4)_{nn} + \sum_{k \neq n} \frac{(K_1)_{nk}(K_3)_{kn} + (K_3)_{nk}(K_1)_{kn} + (K_2)_{nk}(K_2)_{kn}}{D_k} \\ & + \sum_{k,l \neq n} \frac{(K_1)_{nk}(K_1)_{kl}(K_2)_{ln} + (K_1)_{nk}(K_2)_{kl}(K_1)_{ln} + (K_2)_{nk}(K_1)_{kl}(K_1)_{ln}}{D_k D_l} \\ & + \sum_{k,l,m \neq n} \frac{(K_1)_{nk}(K_1)_{kl}(K_1)_{lm}(K_1)_{mn}}{D_k D_l D_m} - \kappa_2 \sum_{k \neq n} \frac{(K_1)_{nk}(K_1)_{kn}}{D_k^2}. \end{aligned}$$

This includes finite- $r$  vertex and frequency corrections,  $V_4^2$ , all three  $V_3^2 V_4$  placements,  $V_3^4$ , and the normalization subtraction. The allowed shifts make every sum finite. Exact simplification reproduces the recursive  $\kappa_2$  and  $\kappa_4$  above identically.

### 4 Conversion to the drive variable

With a separate  $\epsilon = \eta^{-1/3}$ ,

$$r = \epsilon^{-1} - \frac{\delta}{3}\epsilon + \frac{\delta^3}{81}\epsilon^5 + O(\epsilon^7),$$

which satisfies  $r^3 + \delta r = \epsilon^{-3} + O(\epsilon^5)$ . Reconstructing

$$\frac{E_n}{V} = E_{\text{vac}}^{(2)} + r^2 \left( \sqrt{3}n + \lambda^2 \kappa_2 + \lambda^4 \kappa_4 \right)$$

before this substitution yields

$$\boxed{\begin{aligned} \frac{E_n}{V} = & -\frac{3}{2}\eta^{4/3} + \left[ \delta + \sqrt{3}\left(n + \frac{1}{2}\right) - 1 \right] \eta^{2/3} \\ & + \frac{\delta(1-2\delta)}{6} - \frac{6n^2 + 6n + 1}{72} + K_n(\delta)\eta^{-2/3} + O(\eta^{-4/3}), \end{aligned}} \quad (1)$$

where

$$K_n(\delta) = \frac{1}{2592} \{ 96\delta^3 + (144\sqrt{3}n - 288 + 72\sqrt{3})\delta^2 \\ + (144n^2 - 288\sqrt{3}n + 144n - 144\sqrt{3} + 312)\delta \\ + 10\sqrt{3}n^3 + (15\sqrt{3} - 144)n^2 + (139\sqrt{3} - 144)n + 67\sqrt{3} - 120 \}. \quad (2)$$

The exact quadratic part contributes

$$\frac{\delta^3}{27} + \frac{\sqrt{3}\delta^2 n}{18} - \frac{\delta^2}{9} + \frac{\sqrt{3}\delta^2}{36};$$

the remainder of (2) is the complete interaction contribution.

The spacing correction is

$$K_{n+1} - K_n = \frac{72\sqrt{3}\delta^2 + (144n - 144\sqrt{3} + 144)\delta + 15\sqrt{3}n^2 + (30\sqrt{3} - 144)n + 82\sqrt{3} - 144}{1296},$$

and the second-difference correction is

$$K_{n+2} - 2K_{n+1} + K_n = \frac{48\delta + 10\sqrt{3}n + 15\sqrt{3} - 48}{432}.$$

## 5 Status

|             |                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Exact       | Displacement identity, Bogoliubov transformation, sign cancellation, ladder actions, and equality of the two finite calculations. |
| Formal      | Fixed- $n$ Rayleigh–Schrödinger expansion through $\eta^{-2/3}$ .                                                                 |
| Conjectural | Its identification with zeros of the global Olver compensated-to-essential coefficient.                                           |
| Open        | A rigorous operator remainder uniform in $n$ , the global canonical domain chain, and the Olver connection-error estimate.        |